

SPONSOR ME, SLAYERS!

A Futuristic Gladiatorial Game Show Twin-Stick Shooter

Author: Kailee Nekoba

1. Executive Summary

Core Concept: A fast-paced, top-down 3D twin-stick arcade shooter built in the Unreal Editor for Fortnite (UEFN) using the Verse programming language. The game decouples character movement from weapon aiming, requiring players to run in one direction while targeting and firing independently using mouse coordinates.

Setting & Narrative Tone: Set in a dystopian, futuristic televised gladiatorial sports broadcast, the game challenges a contestant to survive waves of hostile cybernetic threats inside a single-room stadium arena. The tone is broad, self-aware game-show comedy in the tradition of Smash TV and Total Carnage. Corporate hostility and glitz are played for laughs through sarcastic on-screen commentary, flashing scoreboard visuals, and physical, over-the-top household appliance prize drops.

Contestant & Enemy Lore: Contestants volunteer for the broadcast to escape crushing financial debt. The Network notoriously never pays survivors because no player has ever completed a run alive—a bleak reality mockingly broadcast by the show's holographic host. The cybernetic threats are built from repurposed, rejected pilot-episode robot models and bankrupt former contestants who accepted mechanical chassis modifications as a severance package.

Core Gameplay Loop: The player destroys cybernetic hostiles to earn cash and Hype. Filling the Sponsor Hype Meter triggers audience-sponsored paraglider supply crate drops containing weapons and shields, enabling the player to destroy scaling waves for exponential financial rewards.

1.1 Look and Feel (Visual & Environmental Design)

- **Visual Perspective:** A locked top-down camera with a bird's-eye orthographic perspective (objects maintain constant scale regardless of position, removing lens distortion). The camera is anchored at a fixed height above the flat stadium floor, providing a complete view of incoming threats.
- **Arena Aesthetics:** A rectangular, industrial stadium bounded by hazard-striped security fences and illuminated by towering, blinking floodlights.
- **Environmental Obstacles:** The stadium floor features static electrical power grids and broken concrete barrier debris. Both hazards block player movement and enemy pathfinding, forcing frantic tactical maneuvering.
- **Visual FX:** Flashing high-contrast neon signs, bright colored paragliders attached to falling crates, and defeated enemies that dissolve into electrical sparks upon destruction.
- **Audio FX:** A satirical game-show buzzer at wave completion, a retro synth-wave soundtrack, and an upbeat, high-pitched 'ding-ding-ding!' sound loop triggered on cash

collection.

- **Art Style & Scope:** Inspired by Paper Boy, Toobin', and Smash TV, using Vampire Survivors as a direct model for low-fidelity, high-readability sprite and tile assets. Characters do not feature complex multi-directional walk animations, restricting the visual scope to a realistic level for a 6-week solo development schedule.

2. Player Experience & Core Mechanics

2.1 The Core Recursive Gameplay Loop: The core loop is structured as a vertical, progressive sequence of mechanical player actions:

- 1. Defeat Hostiles & Collect Cash:** The player destroys cybernetic threats, causing them to burst into a physical shower of coins, cash bundles, and retro household appliances (toasters, TVs) with physics simulation.
- 2. Surge HUD Hype Meter:** Collecting physical drops and performing high-skill combat maneuvers (such as close-shave dodges) fills the vertical HUD Hype Meter.
- 3. Trigger Paraglider Drops:** Crossing Hype thresholds prompts the simulated, televised streaming audience to parachute supply crates directly into the arena.
- 4. Equip Weapons & Upgrades:** Touching a crate instantly equips specialized weapons or shields into the player's active slots, replacing standard gear.
- 5. Survive Escalating Waves:** The player deploys their upgraded arsenal to survive denser, faster, and more aggressive waves of cybernetic threats.
- 6. Accumulate Exponential Bankroll:** Clearing room waves yields massive cash windfalls, which save locally to increment the player's persistent cosmetic rank.

2.2 Controls & Twin-Stick Aiming

Movement is controlled using the standard WASD keys in eight directions. Weapon aiming is entirely independent, tracked dynamically via the mouse cursor. This twin-stick structure allows players to run in one direction while continuously firing in another, enabling key circular 'kiting' and evasion techniques.

2.3 Cash & Retro Prize Drops ('Ding-Ding-Ding!')

Hostiles burst into a shower of physical loot: gold coins, cash bundles, and nostalgic household appliance props (such as retro TVs and toasters). Walking over these props triggers an immediate collision pickup, playing an upbeat 'ding-ding-ding!' audio stinger and incrementing the player's run score.

2.4 First-Life Onboarding Ramp

To flatten the learning curve and prevent HUD clutter, systems are introduced incrementally over the first five room-loops on the player's first life:

- **Room-Loop 1:** Basic WASD movement, independent mouse aiming, standard Pulse

Blaster weapon, and weak melee Cyber-Swarmers only. HUD indicators, Hype systems, and crates are deactivated.

- **Room-Loop 2:** Unlocks basic weapons crates, physical cash drops, the vertical Hype Meter, and the manual Hype Call ability.
- **Room-Loop 4:** Integrates tiered crate drops (Underdog, Rising Star, Superstar) that scale directly with active Hype.
- **Room-Loop 5:** Unlocks the simulated live-stream chat feed widget on the left side of the HUD.
- **Restart Skip:** Upon death, restarting immediately skips this ramp. The next run launches on loop 1 with all systems and tiered crates active from the start.

2.5 Win/Loss Resolution & Loop Termination

- **Win State (Room Won):** A room is cleared when the player successfully eliminates all spawned waves in the active Escalation Tier. A loud game-show buzzer sounds, environmental coordinates and concrete obstacles reset, and the next Escalation Tier begins.
- **Loss State (Run Lost):** A run ends permanently when the player's health pool is depleted to zero and they fail to collect a spawned Sponsor Aid turkey leg before the 3-second 'Death Save' countdown timer expires.

2.6 Career Sponsor Rank (Persistent Progression)

To incentivize repeat play sessions, a lightweight, locally-saved statistic tracks the player's Career Sponsor Rank. At run termination, the game compares the final score and highest tier reached against the player's saved records. Beating either threshold advances the rank through five sarcastic titles: Debt-Ridden Rookie, Undercard Filler, Fan Favorite, Ratings Magnet, and the Network's Sweetheart. Advancing unlocks a cosmetic holographic host title card on the main menu and triggers a custom commentator bark at match start. Career Rank is purely cosmetic and has zero impact on combat math or enemy difficulty, making it extremely lightweight to engineer.

3. Simulated Audience & Upgrade Mechanics

3.1 Sponsor Hype Meter

The vertical, thermometer-style HUD bar tracks player style. Rapid multi-kills, close-shave dodges, and prize pickups generate Hype. If health drops below 40%, the Underdog Boost activates, granting +50% Hype generation. The meter decays by 5% every 10 seconds of inactivity. Pressing and holding the Hype Call key for 1 second triggers a manual Hype Call (10-second cooldown), granting an instant burst of Hype. Active Hype levels determine the quality tier of falling paraglider supply crates.

3.2 Color-Coded Paraglider Crates

Supply crates parachute from the stadium ceiling, displaying high-contrast colored

paragliders that denote their quality tier. Crates trigger instantly upon player collision to maintain momentum. The player character features four upgrade slots: Weapon, Consumable, Shield, and Ammo Modifier. Picking up a duplicate upgrade refreshes its active duration rather than stacking, preventing runaway stat exploits.

3.3 Weapon & Boost Profiles

Weapon / Boost	Visual & Audio Behavior	Tactical Answer & Gameplay Effect
Standard Pulse Blaster	Moderate fire rate; fires streams of glowing blue plasma bullets; distinct sci-fi ping.	Default infinite ammo weapon. Reliable fallback option when special weapons run out of ammo.
Submachine Gun (Weapon)	Rapid parallel yellow laser fire; heavy mechanical shredding sound. Consumes a 50-round magazine with a 1.2s reload.	BLEED STATUS: Inflicts a bleed effect dealing 5 damage/second over 3 seconds. Ideal for melting low-health Cyber-Swarmers.
Shotgun (Weapon)	Wide horizontal 5-pellet red-mist spread; explosive sonic blast. Consumes a 5-shell tube with a 2.5s shell-by-shell reload.	CHAIN KNOCKBACK: High-impact blasts stagger enemies and launch armored Cyber-Boars backward into other hostiles.
Sponsor Sniper (Weapon)	Slow-charging, bright red laser-sight beam; sharp sonic crack. Consumes a 1-round chamber with a 3.0s bolt-action reload.	PIERCING BEAM: Emits an overcharged energy beam that pierces through lines of distant, stationary Ranged Sentinels.
Sponsor Aid (Consumable)	Parachuting golden roasted turkey leg mesh; bright green pulsing visual overlay.	INSTANT HEAL: Restores 25% of the contestant's maximum health pool immediately upon collision pickup.
Sponsor Aegis (Shield)	Bright pink, translucent hexagonal energy bubble wrapping the player character.	DAMAGE BLOCK: Absorbs up to 3 hostile hits, enabling safe, aggressive crowd-ramming maneuvers to escape traps.

Flaming Ammo (Modifier)	Incendiary projectile modifications leaving glowing orange neon trails.	TICKING BURN: Layers ticking burn damage on top of SMG, Shotgun, or Sniper shots. Highly effective against heavy elite tanks.
Icy Rounds (Modifier - Stretch Goal)	Frosted projectile modifications leaving pale-blue crystalline trails; sharp ice-crack sound.	SLOW FIELD: Layers a stacking movement-speed debuff (-20% per stack, max 3) onto hit targets. Used to kite dense melee swarms.

3.4 The 'Death Save' Window & Rescue Methods

Taking a fatal blow triggers a 3-second 'Death Save' slow-motion countdown. Gameplay slows to a crawl, the screen desaturates to grayscale, and the simulated streaming chat floods with supportive text as the commentator screams for a sponsor rescue. To survive, the player must secure a Sponsor Aid item before the timer hits zero via:

- **The Hype Call (Rescue Teleport):** Taking a fatal blow instantly resets the player's Hype Call cooldown. Pressing the key initiates a Hype-scaled rescue teleport. Success rates scale with Hype quality: 35% at Underdog, 50% at Rising Star, and 65% at Superstar. On success, a Sponsor Aid item spawns directly at the player's feet.
- **The Manual Run:** If the Hype Call fails or is skipped, a physical Sponsor Aid turkey leg always spawns on the stadium floor within easy walking distance. The player can manually run and collect it to revive.
- **Anti-Chain Exploits:** The Death Save is restricted to once per life. A second fatal blow taken before the player's health regenerates above 25% results in instant run termination.

3.5 Simulated Audience & Chat HUD (Stretch Goal / Backlog)

To capture the high-stakes game-show environment offline, a scrolling simulated stream chat widget is positioned on the left side of the HUD. AI-driven bot profiles react to combat spikes and trigger drops. To protect player visibility during high-intensity close-quarters combat, the chat feed dynamically fades to 30% opacity whenever hostiles are within immediate melee range of the player character.

4. AI Architecture & Agent Roles

To deliver a highly polished, functional prototype on a compressed schedule, development work is distributed across four specialized AI coding agents. Each agent acts as a virtual studio department under the direct supervision and review of the human designer, ensuring

codebases remain clean, scalable, and modular.

Agent Name	Core Coding Responsibility & Output	UX Outcome (What the Player Sees/Feels)
Gameplay Systems	Scripts character movement physics, WASD inputs, independent mouse aiming vectors, weapon firing cycles, collision-based prize pickup, and wave spawner loops in Verse (.vs).	Smooth, responsive character movement and aiming. Seamless weapon swaps and scaling wave pressure.
Simulated Audience	Develops the HUD Hype Meter bar widget, the simulated streaming chat interface, and handles random-coordinate crate spawn parameters in Verse (.vs).	An immersive, responsive, and dynamic broadcast atmosphere that instantly rewards stylish play.
Announcer Bark	Structures and compiles the sarcastic game-show commentator dialogue database in Verse (.vs), mapping barks to triggers. Creative dialogue is written manually by the human.	Real-time verbal commentator feedback that mockingly calls out player deaths or celebrates multi-kills.
Playtest QA	Parses raw output logs (qa_log.txt) and code diffs from manual playtests using a free, locally-hosted Llama-3 model to isolate navigation-mesh errors and crash vectors.	A locked 60 FPS gameplay experience entirely free of collision traps, memory leaks, or pathfinding deadlocks.

4.1 Development-Time Coordination Cycle (Human-in-the-Loop)

To avoid uncontrolled development costs and code integration drift, the four AI agents are strictly prohibited from communicating or prompting one another directly. The human designer acts as an absolute checkpoint between handoffs. When an agent completes a development ticket, the output is manually audited, approved, and integrated by the designer. Only after human verification does the designer copy and paste the relevant interfaces or data schemas into the next agent's prompt, establishing a reliable human-in-the-loop checkpoint system.

4.2 Runtime Subsystem Communications & Data Flows

Once compiled, the game's runtime systems communicate automatically to pass events

and state changes. The following table maps the exact data exchange pathways established by the development team:

Sender System	Receiver System	Payload / Event	Active Gameplay Result
Gameplay Systems Agent	Simulated Audience Agent	CombatTelemetryEvents	Streams multi-kills, dodges, and cash pickups to fill the vertical Hype Meter widget.
Simulated Audience Agent	Gameplay Systems Agent	SpawnCrateEvent	Sends Hype quality tier (Underdog, Rising Star, Superstar) and 3D location to spawn paraglider drops.
Simulated Audience Agent	Announcer Bark Agent	#DeathSaveTriggered	Triggers a specific sarcastic barks database lookup to fire real-time commentator voice lines.
Playtest QA Agent	Gameplay Systems Agent	Code Patch Diffs	Pipes automated object-pooling optimization parameters to Systems based on qa_log.txt bug patterns.

5. Technical Strategy & Constraints

5.1 Verse Concurrency Performance

To prevent processing bottlenecks during intense physics, visual FX, and UI updates, the Verse codebase decouples core movement and collision handling from visual assets. Secondary visual effects, particle streams, and HUD counters are executed asynchronously via Verse structured concurrency using spawn and branch expressions, guaranteeing a locked 60 FPS performance envelope.

5.2 Announcer Latency Guardrail

Streaming dialogue files dynamically over the internet can cause audio clipping and latency, ruining the comedic timing of the host's barks. To resolve this, all 25 pre-written barks are

loaded into the game memory at initial runtime, permitting instantaneous, delay-free audio triggers from local storage.

5.3 Rigid Object Budget & Despawn Rules

To remain within UEFN's strict ~100-active-prop platform memory limit, the wave spawner caps concurrent active hostiles at 40 bots. Spawns are blocked within a 3-meter safety radius of the player to prevent instant collision damage. Bullet props, cash bundles, and expired visual FX are aggressively culled and despawned exactly 5 seconds after generation.

5.4 Minimum Playable Product (MVP) Target Quantities

To ensure successful shipping by the capstone deadline, the project commits to strict asset ceilings:

- **Environment:** 1 rectangular industrial stadium arena, 2 static obstacle types (electrical grids, broken concrete).
- **Characters:** 1 playable contestant model, 2 cybernetic hostile models (melee Swarmer, heavy Ranged Tank).
- **Weapons:** 1 standard Pulse Blaster default sidearm, 3 crates weapons (SMG, Shotgun, Sniper), 4 boost profiles.
- **Audio Assets:** 1 retro synth-wave music track, 2 game-show buzzer sound effects, 25 compiled announcer bark audio strings.
- **HUD Widgets:** 1 vertical Hype Meter bar, 1 Death Save slow-motion overlay, 1 simulated streaming text chat widget.

5.5 Difficulty Scaling & Escalation Curve

Completing a room cleared wave escalates difficulty by exactly 8% per tier. Hostile maximum health pools, movement speeds, and spawn densities scale up incrementally. To prevent late-game scenarios from becoming completely unplayable, scaling is hard-capped at Escalation Tier 21 (~5x starting difficulty), maintaining a highly challenging but fair late-game.

5.6 Development Timeline & Academic Milestones

- **Week 1 (Jul 22–28):** Establish core WASD player movement, mouse aiming vectors, and standard Pulse Blaster in gray-box room.
- **Week 2 (Jul 29–Aug 4):** Integrate SMG, Shotgun, and Sniper crates with timer overrides; establish physical cash loot drops and basic Hype meter.
- **Week 3 (Aug 5–11):** Write wave difficulty scaling curves, enforce the 40-enemy concurrent limit, and implement Sponsor Aid healing drops.
- **Week 4 (Aug 12–18):** Build tiered crate qualities (Underdog, Rising Star, Superstar) and locally persistent Career Sponsor Rank tracking.
- **Week 5 (Aug 19–25):** Write and map the 25 sarcastic announcer audio barks; set up simulated stream chat HUD; run Playtest QA parses.
- **Week 6 (Aug 26–Sep 1):** Perform in-editor 'Launch Session' playtesting, clean up

nav-mesh collision paths, freeze systems, and ship.

5.7 Plausible Token Budget & Scope Cuts

To maintain zero-cost development, the QA Agent log parser utilizes a free, locally-hosted Llama-3 model. Cloud API tokens are reserved for compiling final Verse code, budgeted at 4,500,000 total tokens (45% Systems, 20% QA, 15% Bark, 10% Audience, 10% Reserve). If schedule slippage occurs, scope cuts resolve sequentially in this exact order:

1. Cut the Simulated Stream Chat HUD widget entirely (announcer barks carry the game-show commentary narrative).
 2. Cut the Icy Rounds modifier (collapsing ammo modifiers to just the Flaming Ammo burn ticks).
 3. Cut the Flaming Ammo modifier entirely (collapsing weapons profiles strictly to the standard sidearm and 3 basic weapons).
 4. Cut tiered paraglider crate scaling (collapsing supply drops into a single, flat un-tiered crate quality).
- The twin-stick aiming controls, wave spawning loops, Win/Loss resolution conditions, and Career Sponsor Rank remain strictly uncuttable.

6. Revision History & Growth Log

This log documents the developmental evolution of the GDD, capturing feedback from peers, advisors, and agent reviews to ensure complete architectural transparency before academic submission:

Feedback Source	Critique & Gaps Identified	Technical Resolution & GDD Update
Chris Porter	Noted that Fortnite is built in Unreal but UEFN utilizes written Verse code instead of standard visual blueprints.	Re-architected the technical strategy, re-scoping deliverables to focus on compiled text-based Verse scripts (.vs files).
Josh Rose	Flagged that combat was dry and felt unrewarding without satisfying arcade-shooter scoring loops.	Added physical, physics-based 3D prize drops (coins, cash, toasters, TVs) that play a high-pitched 'ding-ding-ding!' stinger.
Chris & Josh	Flagged the original duplicate-weapon 'Overcharge' and 'spacebar-smash' crates as	Simplified crate interactions to instant-collision triggers and added a Flaming Ammo modifier that layers onto active

	clunky and overly complex.	weapons and refreshes on pickup.
Agent QA	Flagged that calling live web text-to-speech audio during wave spawning caused game stutter and dialogue clipping.	Implemented the Announcer Latency Guardrail, pre-loading sound files and reading barks locally from a compiled Verse database module.
Six-Agent Board	Flagged conflicting health rules (Loss vs. Death Save), infinite scaling bugs, and a lack of persistent player progress.	Reconciled Death Save/Loss conditions; established a 40-enemy cap, 3m safety spawn radius, same-slot refresh, and persistent cosmetic Career Rank.
Team Debrief	Flagged that the GDD omitted the core humorous game-show tone, reading like a dry military tactical shooter.	Added an explicit tone declaration to the Executive Summary (broad, self-aware game-show comedy in the tradition of Smash TV).
Chris Porter	Advised that each weapon suggest different player behavior and that having only two weapons felt thin.	Added the precision-based Sponsor Sniper weapon for spacing play, and added status effects (SMG bleed, Shotgun chain knockback).
Cap Review Pass	Noted lack of explicit inter-agent communication data flows and missing hard counts for MVP scope control.	Added Section 4 Inter-Agent Communication & Data Handoffs workflow, mapped agent UX outcomes, and defined Section 5 MVP Target Asset Quantities.
Josh Rose	Flagged that sarcastic announcer dialogue must be hand-written by the designer, with AI limited to brainstorming assistance.	Reworded the Announcer Bark Agent's responsibility to reflect human-authored lines, restricting the agent to database structuring.
Kai	Career Sponsor Rank was named as an uncuttable feature in Timeline and Scope Cuts but was never defined in player-facing mechanics.	Added a Career Sponsor Rank subsection (Section 2) defining it as a lightweight, cosmetic-only rank ladder tied to best-run cash/Escalation Tier.

Chris Porter	Requested separate Look & Feel bullets, plain-English game loop teaser up front, removal of Executive Summary tech fluff, explicit win/loss conditions.	Confirmed Look and Feel bullets in Section 1; added plain Core Loop line to Executive Summary; moved technical details to Section 5; added explicit win/loss conditions.
Chris Porter	Said the GDD still didn't let a reader picture the game's look. Pointed to Paper Boy, Toobin', and Smash TV as references.	Added an Art Style & Scope bullet to Section 1 naming references and committing to simple, repeated tile art over custom animation.
Chris Porter	Noted that AI agents should have a human checkpoint between handoffs instead of self-prompting one another.	Added a Development-Time Coordination Cycle subsection (Section 4) describing the designer-mediated accept/reject handoff between agents.